

PRASANJIT DUBEY

Ph.D. Candidate, Machine Learning, Georgia Tech · Graduating May 2027

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Education

Georgia Institute of Technology

Expected May 2027

H. Milton Stewart School of Industrial and Systems Engineering (ISyE)

Ph.D. in Machine Learning, Minor in Mathematics | Advisor: Prof. Xiaoming Huo | GPA 3.85/4.0

Indian Institute of Technology, Bombay

2017 – 2019

M.Sc. in Applied Statistics & Informatics | Department Rank 1/38 | GPA 8.96/10

University of Calcutta

2014 – 2017

B.Sc. (Honors) in Statistics, Minor in Mathematics & Computer Science | Grade 92.5%

Professional Experience

Bloomberg L.P.

Jun 2025 – Aug 2025

Quant Intern, Quantitative Risk Analytics (Enterprise Products), New York, USA

- Built a **Kalman-filter hazard-curve model** for pricing credit instruments by estimating dynamic CDS term structures from **latent Nelson–Siegel factors and macroeconomic covariates**, **integrated into Bloomberg Terminal’s Credit VaR engine** to forward-simulate risk-neutral hazard rates.
- Implemented a **Credit Rating Migration module** for the Credit VaR engine using **macro-conditioned multinomial-logit and intensity-based models** to simulate obligor- and index-level rating paths, capturing systematic and idiosyncratic risk.
- Designed a **hybrid correlation model** combining economic factor loadings with **PCA-calibrated residuals** to simulate joint firm-value shocks across obligors.

AT&T Labs, Inc.

May 2024 – Aug 2024

Research Scientist Intern, Data Science & AI Research, New Jersey, USA

- Developed a novel framework for analyzing **sparse longitudinal customer-journey text** by integrating **large language models (LLMs)** with **multivariate functional data analysis**.
- Applied **sparse multivariate functional PCA** with user covariates for segmentation, **anomaly detection**, and recovery of key modes of variation in the customer journey.
- Used **LLM-guided dynamic keyword profiling** on real-world data (Amazon reviews) to surface customer pain points and actionable insights, **outperforming existing customer-journey analytics**; shipped a **production application** (Emotional AI) for behavioral anomaly detection and targeted interventions.

Australia & New Zealand (ANZ) Banking Group

Jun 2019 – Aug 2021

Senior Quantitative Analyst, Wholesale Credit Risk Modeling, Bengaluru, India

- Built the **Financial Ratio Surveillance Tool (FRST)**, a production **R Shiny** stress-testing application to forecast and analyze how effectively financial ratios predict the probability of default.
- Developed a **multivariate financial-ratio analytics suite** (sigmoid calibration, kernel regression, shape-constrained additive models, spline smoothing) paired with a novel **branched forward-selection** routine to auto-select the top- n credit-risk models across diverse portfolios.
- Led the bank’s **statistical COVID-19 impact assessment** across commercial portfolios, flagging high-volatility sectors for stakeholders as part of **APRA regulatory** requirements.

Walmart Labs

May 2018 – Aug 2018

Research Intern, Data Science (International Pricing & Demand Forecasting), Bengaluru, India

- Devised an **ensemble of ridge, lasso, and elastic-net models** on high-dimensional, large-scale data to forecast sales across product categories using item-, region-, and covariate-specific **price elasticities**.
- Built a **cluster-based (k -means) price-elasticity model**, **reducing demand-forecast error by 42.4% (MAPE)** over Walmart’s incumbent methods.

Research Papers

- [1] **Anytime-Valid Federated Conformal RAG for LLM Swarms.** P. Dubey & X. Huo. *Under review at Conference on Neural Information Processing Systems (NeurIPS), 2026.* [\[Paper\]](#)
- [2] **Matching Rates and Optimal Allocation for Federated Probe-Logit Distillation under Heterogeneous Bandwidth Budgets.** P. Dubey & X. Huo. *Under review at Statistics and Trustworthy AI for Cross-Domain Acceleration (STAI-X), Harvard, 2026.* [\[Paper\]](#)
- [3] **Federated Language Models under Bandwidth Budgets: Distillation Rates and Conformal Coverage.** P. Dubey & X. Huo. *Under review at Conference on Neural Information Processing Systems (NeurIPS), 2026.* [\[Paper\]](#)
- [4] **BOOST: Power-Optimal Strong-FWER Testing for Block-Structured Multiplicity.** P. Dubey & X. Huo. *Under review at Conference on Neural Information Processing Systems (NeurIPS), 2026.* [\[Paper\]](#)
- [5] **Optimal Multiple Testing under Family-Wise Error Control: Elementary Symmetric Polynomials and a Scalable Algorithm.** P. Dubey & X. Huo. *Under review at Biometrika, 2026.* [\[Paper\]](#)
- [6] **LLmFPCA-detect: LLM-powered Multivariate Functional PCA for Anomaly Detection in Sparse Longitudinal Texts.** P. Dubey, A. Guha, Z. Zhou, Q. Wu, X. Huo, & Paromita Dubey. *Under review at IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), 2025.* [\[Paper\]](#)
*2nd Place Paper Award, Student Research Showcase, 2025 INFORMS Regional Analytics Conference (IRAC)
- [7] **Most Powerful Test with Exact Family-Wise Error Rate Control: Necessary Conditions and a Path to Fast Computing.** P. Dubey & X. Huo. *Under review at Journal of Computational and Graphical Statistics (JCGS), 2025.* [\[Paper\]](#)
*Invited talk, Optimisation for Statistics and Learning session, 2025 INFORMS Annual Meeting
- [8] **Generalization Bounds for Magnitude-Based Pruning via Sparse Matrix Sketching.** E. K. Guha, P. Dubey, & X. Huo. *International Conference on Learning Representations (ICLR) Workshop, 2024.* [\[Paper\]](#)

Technical Skills

- **Programming:** Python, C++, R, R Shiny, SQL, MySQL, MATLAB
- **ML Frameworks:** LLMs, PyTorch, scikit-learn
- **Computing & Tools:** High-Performance Computing (Slurm), GPU computing, Apache Hadoop (MapReduce, Streaming), Git
- **Graduate Coursework:** High-Dimensional Probability & Statistics, Statistical Inference, Time Series Analysis, Stochastic Processes, Statistical Techniques in Data Mining, Convex & Stochastic Optimization, Computational Data Analysis

Awards & Honors

- **2nd Place Paper Award**, Student Research Showcase, INFORMS Regional Analytics Conference (IRAC), Atlanta 2025
- **Stewart Topper Fellowship Award**, Georgia Tech 2021 – 2022
- **Dr. P. V. Sukhatme Memorial Award** (M.Sc. Rank 1), IIT Bombay 2019
- **All-India Rank 26**, Graduate Aptitude Test in Engineering (GATE), Statistics 2019
- **Pfizer Scholarship** (M.Sc. first-year Rank 1) 2018
- **All-India Rank 7**, IIT-JAM Mathematical Statistics 2017
- **Swami Asaktananda Award** (highest B.Sc. academic distinction) 2017
- **Summer Research Fellowship**, Indian Academy of Sciences 2016, 2017

Conference Reviewing & Teaching

- **Reviewer:** ICLR, ICML, and NeurIPS 2023 – present
- **Teaching Assistant**, Georgia Institute of Technology 2021 – 2025
 - **Data Mining and Statistical Learning** (graduate-level, 400+ students)
 - **Statistical Modeling and Regression Analysis** (graduate-level, 450+ students)